"Vison Talk"

Gregory Livingston

Final Project *-Computer Vision (ITAI 1378)*

ITAI- 1378

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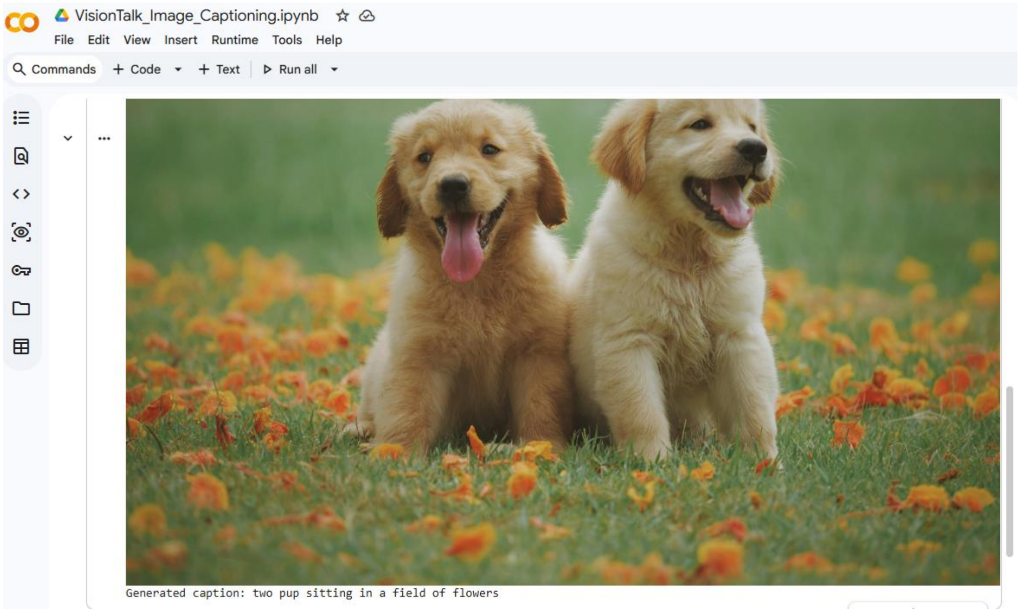


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Generated caption: two pup sitting in a field of flowers
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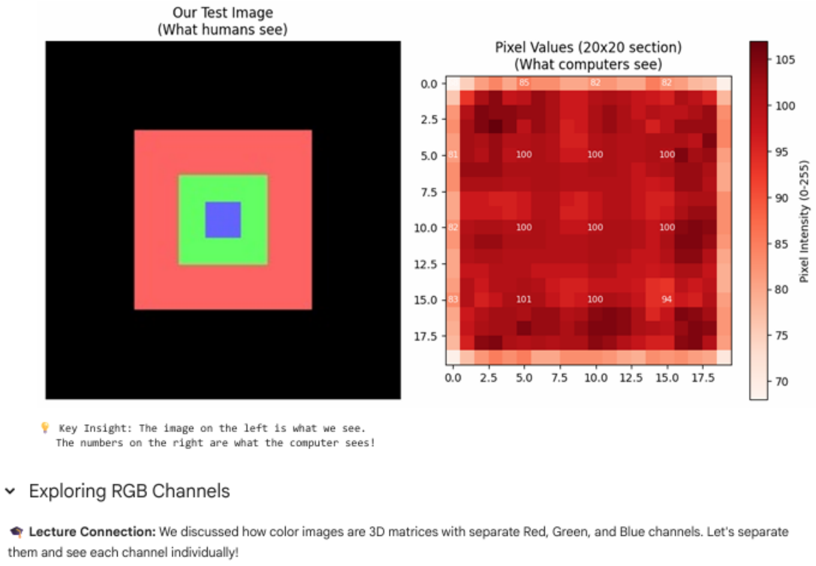
Gregory-Livingston-Applied-AI-Portfolio Public

Portfolio Link: https://github.com/GregLivin/Gregory-Livingston-Applied-AI-Portfolio

What "I" Learned

- I learned how computers interpret images using pixel values and color channels.
- I practiced applying image processing techniques such as filtering, edge detection, and thresholding.
- I learned how to use classic machine learning for image classification before moving into deep learning.
- I studied neural networks and how they learn visual patterns.

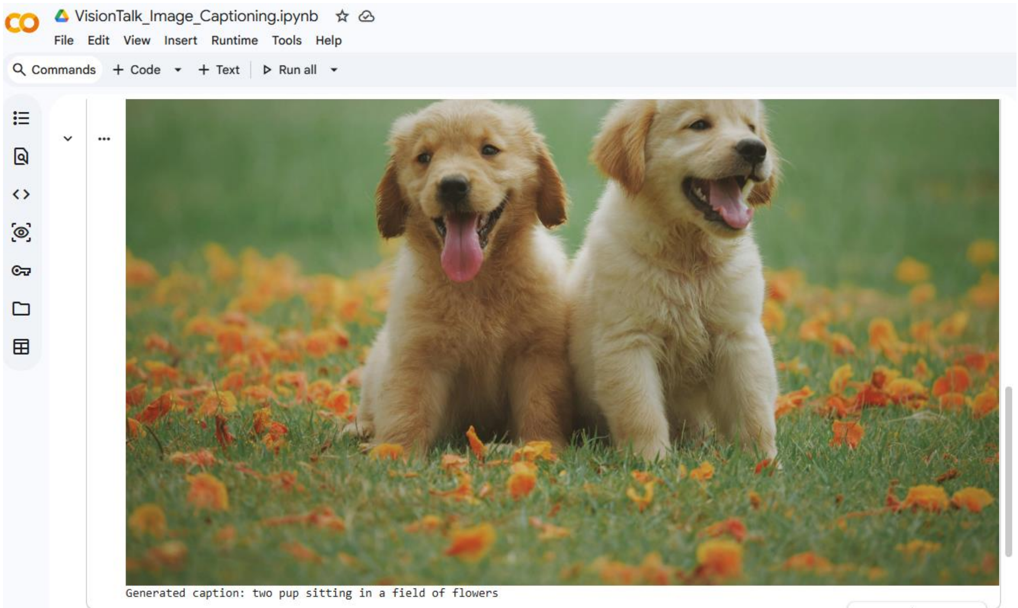


- I learned how Convolutional Neural Networks (CNNs) extract features for image recognition.
- I explored transfer learning and pretrained models for improving performance.
- I learned about Vision Transformers and how attention improves image understanding.
- I studied Vision–Language Models (VLMs) and how they connect images to natural-language generation.

Problem & Solution

Problem:

Images contain a lot of visual information, but without captions, people and systems cannot easily understand the meaning or context.

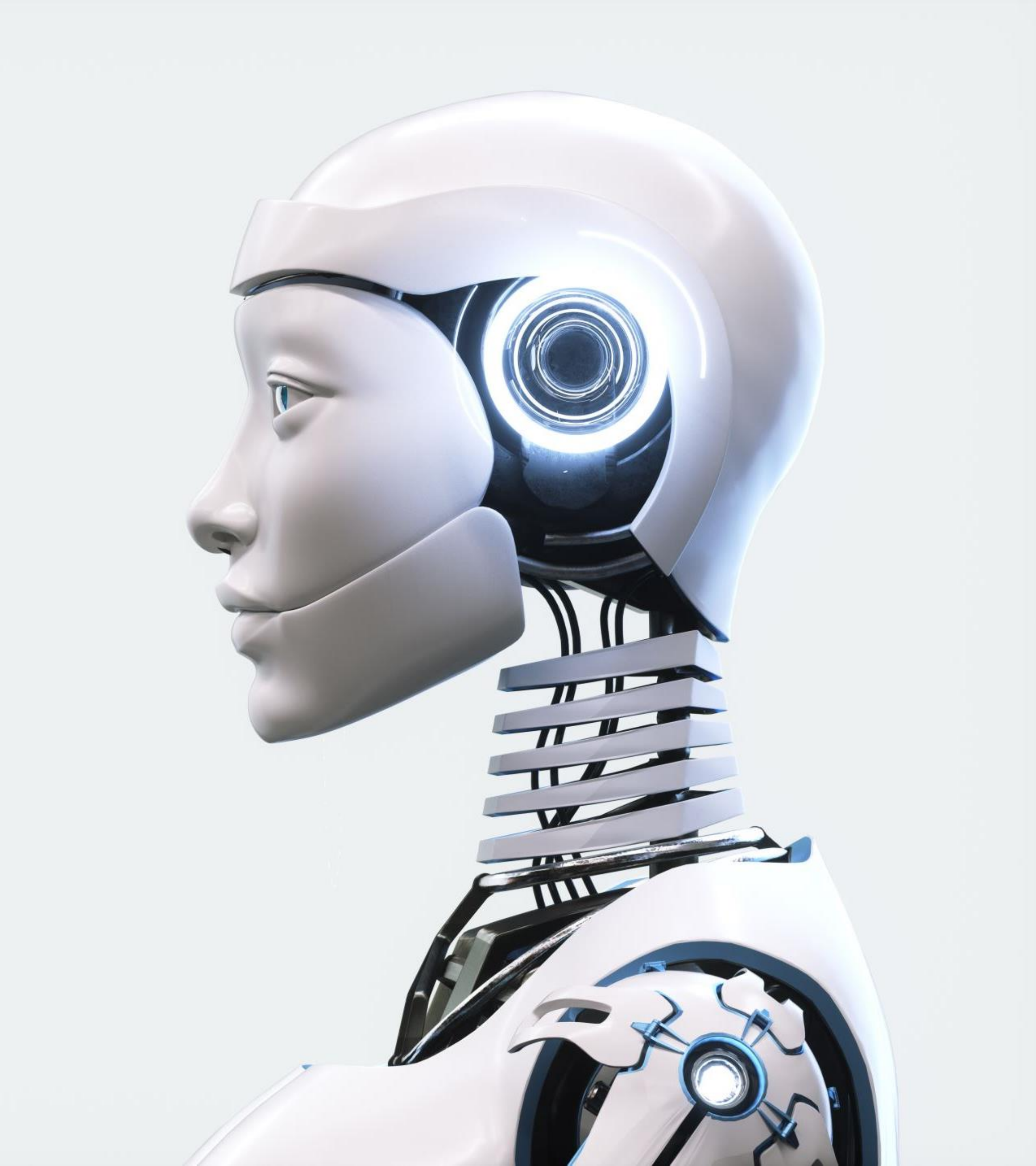


Solution:

I built *VisionTalk*, an AI system that uses a pretrained BLIP Vision–Language Model to automatically generate captions from images. This demonstrates how computer vision can connect visual recognition with natural-language processing.

How the System Works

- Uses a pretrained BLIP model from Hugging Face.
- Takes an input image and encodes visual features.
- The language model generates a natural-language caption describing the image.
- Implemented in a Jupyter Notebook with Python and Transformers.
- Results show meaningful and accurate captions.



What I Gained from This Project

- Hands-on experience using Vision–Language Models
- Improved understanding of how AI connects images and text
- Practice organizing and documenting a full AI project
- Stronger coding and notebook development skills
- Added a complete, functional project to my professional AI portfolio